

iADAS

Intelligent Anomaly Detection and Alerting System

Project Purpose & Objectives



PROJECT PURPOSE

- To build an AI-powered anomaly detection and alert system for an e-commerce business.
- Automatically monitor sales and operational metrics (iPhone, AirPods, iWatch).
- Identify unusual business patterns and generate intelligent business alerts.

PROJECT OBJECTIVES

Detection & Monitoring

Detect anomalies in revenue, orders, refunds, traffic, and spend.
Reduce manual dependency.

Root Cause Analysis

Correlate multiple metrics to identify probable root causes of business anomalies.

Decision Support

Generate plain-language AI alerts to support faster and better business decision-making.

Scope



IN SCOPE

- Analysis of historical e-commerce sales and operational data.
- Monitoring of products such as iPhone, AirPods, and iWatch.
- Detection of anomalies in revenue, orders, refunds, traffic, and spend.
- Correlation analysis between multiple business metrics.
- AI-generated plain-language alerts and summaries.
- Severity-based alert prioritization and reporting dashboard.

OUT OF SCOPE

- Real-time streaming data integration.
- Production-level cloud deployment.
- Integration with external payment, ERP, or logistics systems.
- Automated corrective actions based on detected anomalies.

Deliverables



Data & Metrics

Historical e-commerce dataset containing daily sales and operational metrics.

AI Anomaly Model

AI model which detects anomaly based on sales data (OpenAI models or APIs).

Alert & Summary

AI-generated plain-language alert and summary system.

Prioritization Logic

Severity-based alert prioritization logic for business efficiency.

Reporting & Results

Final project report and presentation demonstrating system architecture, methodology, and results.

Timeline



Review 1 – Project Charter

Saturday, June 13, 2026

- Problem Statement & Objectives
- Scope & Deliverables

Review 2 – Progress Review

Saturday, 04 July 2026

- Dataset & Data Preparation
- AI Model Development
- Anomaly Detection

Review 3 – Final Draft

Saturday, 01 August 2026

- Correlation Analysis
- AI Alert Generation

Review 4 – Final Presentation

Sunday, 16 August 2026

- Testing & Final Enhancements
- Demo & Presentation

Roles & Responsibilities



Vijay Mohan Boddu	Project Lead	Project planning, coordination, stakeholder management
Akash Srivastava	Solution Architect	Solution design, architecture, technology selection
Praveen Kumar Kanni	Solution Architect	Model design, implementation, technical support
Sonal Nesargi	Data Lead	Data collection, cleaning, preprocessing, EDA
Shreyanshu Sheel Jaiswal	Testing Lead	Testing, validation, quality assurance
Sameena Begum	Documentation Lead	Documentation, reporting, presentation preparation

Resources



Software & Tools

- ChatGPT/Co-pilot
- GitHub for version control

Data Resources

- Synthetic sales dataset
- 12 months of metric data
- 2–4 business KPIs (Revenue, Volume, AOV, etc.)

Team Resources

- Multi-disciplinary team for processing, dev, and testing
- Faculty mentor and project guide feedback

Success Criteria



The project will be considered successful if it can achieve the following objectives:

Anomaly Detection

Detect anomalies accurately in selected business metrics.

Severity Classification

Assign meaningful severity levels (Low, Medium, High, Critical).

Root-Cause Analysis

Generate plausible root-cause hypotheses using related metric behavior.

Intelligent Alerting

Produce clear, business-friendly alerts using an LLM.

Validation Integrity

Achieve low false-positive rates during validation.



Thank You
